

Book I. Prop. XLIV. Prob.



To a given straight line (—) to apply a parallelogram equal to a given triangle (▼), and having an angle equal to a given rectilinear angle (▲).

Make $\text{yellow parallelogram} = \text{red triangle}$ with $\text{blue triangle} = \text{yellow triangle}$ (pr. I.42)
and having one of its sides conterminous with and in continuation
of —.

Produce — till it meets — ||;
draw — produce it till it meets continued;
draw || meeting — produced and produce

$\text{yellow parallelogram} = \text{blue parallelogram}$ (pr. I.43)
but $\text{yellow parallelogram} = \text{red triangle}$ (const.)

$\therefore \text{blue parallelogram} = \text{red triangle}$;

and $\text{blue triangle} = \text{red triangle} = \text{black triangle} = \text{yellow triangle}$ (pr. I.29, const.).

Q. E. D.